

Kinematics Solutions

Question 1 (Conceptual)

The displacement s for an object starting from rest ($v_0 = 0$) and accelerating uniformly is given by the kinematic equation:

$$s = v_0 t + \frac{1}{2} a t^2 \implies s = \frac{1}{2} a t^2$$

Calculate the distance for Car A:

- Acceleration (a_A): 2 m/s^2
- Time (t_A): 10 s

$$s_A = \frac{1}{2} (2 \text{ m/s}^2) (10 \text{ s})^2$$

$$s_A = (1 \text{ m/s}^2) (100 \text{ s}^2)$$

$$s_A = 100 \text{ m}$$

Calculate the distance for Car B:

- Acceleration (a_B): 4 m/s^2
- Time (t_B): 5 s

$$s_B = \frac{1}{2} (4 \text{ m/s}^2) (5 \text{ s})^2$$

$$s_B = (2 \text{ m/s}^2) (25 \text{ s}^2)$$

$$s_B = 50 \text{ m}$$

Comparison: Since $s_A = 100\text{ m}$ and $s_B = 50\text{ m}$, **Car A** travels farther. The distance traveled is proportional to the acceleration (a) and the square of the time (t^2). In this case, the longer time Car A spends accelerating has a greater impact on the final distance than the higher acceleration of Car B over a shorter time.

Question 2 (Calculation)

Given values:

- Initial velocity (v_0): 0 m/s (starts from rest)
- Final velocity (v): 20 m/s
- Distance (s): 200 m

(a) Identify the appropriate kinematic equation.

To find the acceleration without knowing the time, the best equation is:

$$v^2 = v_0^2 + 2as$$

(b) Calculate the cyclist's acceleration.

Rearrange the equation to solve for acceleration (a): $a = \frac{v^2 - v_0^2}{2s}$

Substitute the known values:

$$a = \frac{(20\text{ m/s})^2 - (0\text{ m/s})^2}{2(200\text{ m})}$$
$$a = \frac{400\text{ m}^2/\text{s}^2}{400\text{ m}}$$

The acceleration is: $a = 1.0\text{ m/s}^2$

(c) Determine the time taken to reach 20 m/s.

Use the equation that relates initial velocity, final velocity, acceleration, and time:

$$v = v_0 + at$$

Rearrange to solve for time (t): $t = \frac{v-v_0}{a}$

Substitute the values:

$$t = \frac{20 \text{ m/s} - 0 \text{ m/s}}{1.0 \text{ m/s}^2}$$

$$t = 20 \text{ s}$$